Project 2 · Machine Learning: From Data to Predictions

Who will subscribe to a term deposit?

Predicting the outcome of a bank’s phone calls before they’re made

Tamim Wahage

The problem

A Portuguese bank calls clients to sell them a term deposit. Most calls end with a no.

What I predict: will this client subscribe? (before the call)

For whom: the marketing team planning the call list

What they do with it: call the likely clients first, skip the ones who'll almost surely say no

The data

41,188 phone contacts, May 2008 – Nov 2010. One row = one call to one client: client info, earlier contacts, the economy at that time.

The traps I found

- **duration** (call length) is only known after the call → dropped (leakage)
- missing values hide as **“unknown”** and **pdays = 999** → kept as a category / replaced by a flag
- 12 exact duplicates → dropped
- the file is sorted by date and the economy columns act like a hidden date

Source: Moro, Rita & Cortez (2014), Bank Marketing, UCI Machine Learning Repository, doi.org/10.24432/C5K306. Licence: CC BY 4.0.

The metric and the baseline

Only **11.3%** say yes. Saying “no” to everyone is **88.7% accurate** and finds nobody.

A missed yes (lost sale) is worse than a wasted call.

So: **F1**, which is only high if the model finds the yes clients *and* is right when it calls.

Baseline: call everyone

0.203

F1 to beat (finds every subscriber, but 89% of the calls are wasted)

What I tried

4 model families, same 5 folds, with and without class weights. Best of each:

Model	F1 (mean ± spread)	Recall
Gradient boosting, balanced	0.476 ± 0.010	63%
Logistic regression, balanced	0.451 ± 0.009	62%
Random forest	0.376 ± 0.003	30%
Decision tree	0.335 ± 0.011	34%
Baseline: call everyone	0.203	100%

Tuning (grid / random search) didn’t beat the default settings. New features: only small changes, kept the simple ones.

Imbalance and the threshold

Three ways to handle the 11% yes, gradient boosting, cross-validated F1:

Method	at threshold 0.5	at its best threshold
No handling	0.365	0.497
Class weights	0.475	0.499
SMOTE	0.426	0.494

What really matters is **not leaving the threshold at 0.5**. I picked **0.60**: the lowest threshold within the fold spread of the best F1, because a missed yes costs more than a wasted call.

The final model on the test set

Test set (used once)

0.519

F1

Cross-validation

0.496

± 0.012 across folds

Baseline (call everyone)

0.203

F1

With **15% of the calls** the model reaches **61% of the subscribers**. Calling everyone: 100% of the calls, and 89% of them are wasted.

Where it goes wrong

The model mostly recognises **the time period**, not the client. Test set, grouped by the economy at the time of the call:

Period (nr.employed)	Clients	Said yes	Called	Found
Worse economy (≤ 5076)	975	46%	96%	98%
Middle (5099)	1,695	14%	17%	50%
Better economy (5191–5228)	5,564	4.5%	0.4%	4%

Also weak: landline clients (finds 33%) and clients never contacted before (finds 48%).

What drives it

Permutation importance (drop in test F1)

Column	Drop
nr.employed (economy)	0.23
cons.conf.idx (economy)	0.05
poutcome (last campaign)	0.02
euribor3m (economy)	0.02
month	0.02

For the bank

When a client is called matters more than **who** they are: in bad economic times many more people took a deposit.

Within a period: past success, a mobile number, being young or retired.

Conclusions and limitations

Future check: trained on older calls, tested on newer ones → F1 **0.398**, worse than calling everyone (0.468).

Would I use it?

- Not as it is for a new campaign
- Use the patterns now: always call past successes, mobile numbers, students and retired
- Don’t call the same client again and again

With another week

- Validate by time, not randomly
- A model without the economy columns
- Real call costs and deposit value from the bank
- Retrain on recent campaigns

Live demo

The saved model (pipeline + threshold 0.60) is loaded and asked about one client, a 67-year-old retired person with a mobile number:

Same client, called…	P(yes)	Decision
in a bad economic period (Oct, low rates)	0.81	call
in a good economic period (Jul, high rates)	0.37	skip
good period, but last campaign was a success	0.69	call

notebooks/03_evaluation_and_interpretation.ipynb · section 5

Thank you

Who will subscribe to a term deposit?

Tamim Wahage · Questions welcome